LM HW6

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1.1 Data on salaries, Y83 and Y84, for chairs of the 50 largest corporations in Chicago are available. We have information about their age, number of shares they hold, total revenues and income. Using a model with percentage increase given from 1983 to 1984 (i.e., 100*(Y84-Y83)/Y83) as the response, and the remaining 4 variables as predictors, check for outliers. Do you feel the variances of the raises are equal? If you feel they are not equal, what transformation could be used to improve matters? Make appropriate plots and give your conclusions. . . 2

2 Q2 **11**

2.1 Fit a model with the octane rating as the response and the other four variables as predictors. Perform a full set of regression diagnostics. Present the important plots, comment on their meaning and, where appropriate, indicate what action should be taken. Since it is easy to produce a very large number of plots, you will need to be selective. Take care to present examples of each major category of diagnostic. It is acceptable to just report the outcome of some plots without actually displaying them, especially if they do not show anything interesting. 11

3 Q3 **20**

3.1 The data for this question gives observations on the acceleration (ACC) of different vehicles along with their weight-to-horsepower ration (WHP), the speed at which they were traveling (SP), and the grade (G; G=0 implies the road was horizontal). 20

1 Q1

1.1 Data on salaries, Y83 and Y84, for chairs of the 50 largest corporations in Chicago are available. We have information about their age, number of shares they hold, total revenues and income. Using a model with percentage increase given from 1983 to 1984 (i.e., 100*(Y84-Y83)/Y83) as the response, and the remaining 4 variables as predictors, check for outliers. Do you feel the variances of the raises are equal? If you feel they are not equal, what transformation could be used to improve matters? Make appropriate plots and give your conclusions.

Read data:

```
dat1 = read.table("salary.txt", header = TRUE)
head(dat1)
```

```
##       Y84     Y83  SHARES     REV    INC AGE
## 1 1481250 1425000  101037 38828.0 1454.8  65
## 2 1239402 1455350 5713459   558.4   29.7  44
## 3 1205181 1057707   18367 12595.0  479.0  60
## 4 1012500  862500  134713  4233.6  108.3  66
## 5  980000  871056    7896  8321.5  234.1  53
## 6  921213  817687   48722  3104.0  402.6  56
```

Model: $\frac{100(Y84-Y83)}{Y83} = \beta_0 + \beta_1 \text{SHARES} + \beta_2 \text{REV} + \beta_3 \text{INC} + \beta_4 \text{AGE} + \epsilon,\ \epsilon \sim N(0, \sigma^2 I)$

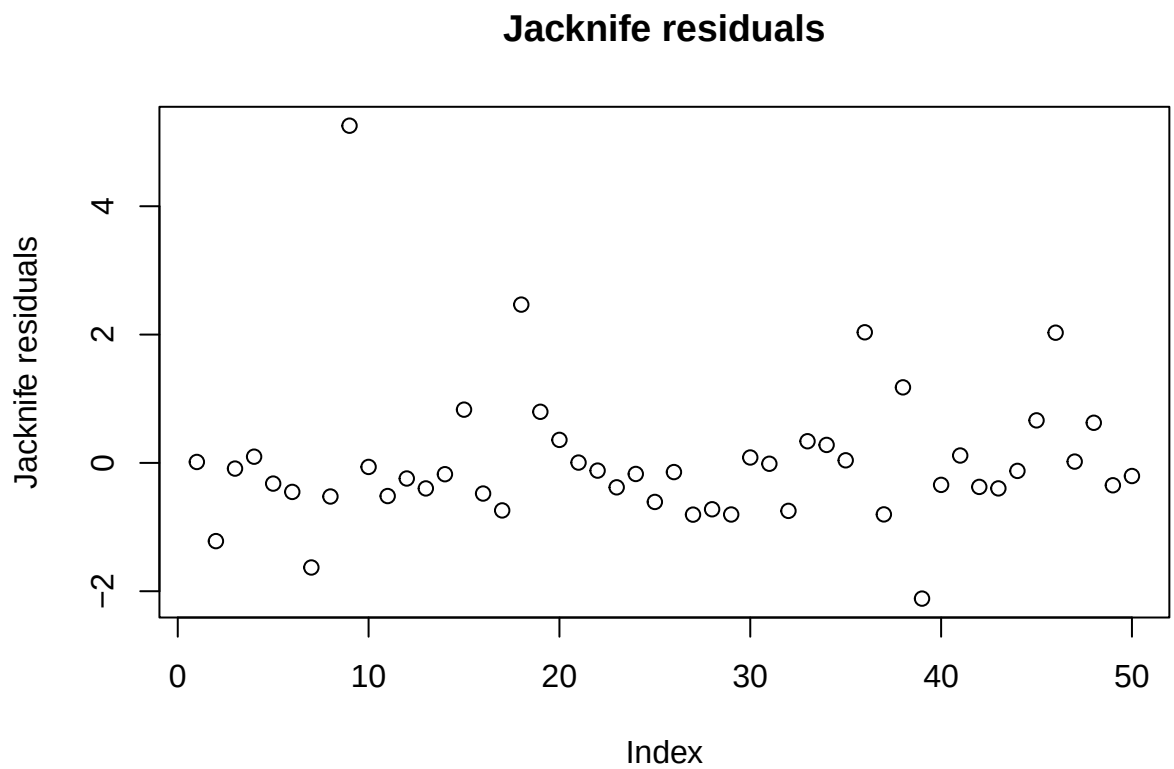
Summary for the fit:

```
fit1 = lm(I(100*(Y84-Y83)/Y83)~., data = dat1)
summary(fit1)
```

```
##
## Call:
## lm(formula = I(100 * (Y84 - Y83)/Y83) ~ ., data = dat1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -53.133 -12.519  -4.066   2.846 109.322
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.509e+01  3.571e+01   1.543    0.130
## SHARES      -3.857e-06  3.717e-06  -1.038    0.305
## REV         -7.237e-04  7.695e-04  -0.940    0.352
## INC          9.744e-03  1.655e-02   0.589    0.559
## AGE         -5.713e-01  6.232e-01  -0.917    0.364
##
## Residual standard error: 26.81 on 45 degrees of freedom
## Multiple R-squared:  0.05754,    Adjusted R-squared:  -0.02623
## F-statistic: 0.6869 on 4 and 45 DF,  p-value: 0.6048
```

Examine outliers

```r
jack = rstudent(fit1)
plot(jack, ylab="Jacknife residuals", main="Jacknife residuals")
```



From the scatter plot above, it is reasonable to suspect that some of the observations might be outliers for their extremely large Jacknife residuals.

It can be shown that Jacknife residuals follow t distribution with degree of freedom $n-p-1$ under null if the model is correct and $\epsilon \sim N(0, \sigma^2 I)$. Hence hypothesis testing described as follows can be conducted to examine if their is any outlier present in this data.

$$\begin{cases} H_0 : \text{No any outlier.} \\ H_1 : \text{There is at least one outlier.} \end{cases}$$

```r
n = nrow(dat1); p = 5
critical = qt(.05/n, n-p-1, lower.tail = FALSE)
```

Note that this proposed test is involved with multiple testing. Bonferroni's correction is one of the measures to deal with such problem. Since the sample size n = 50 and the number of parameters p = 5, we have the critical value for this test being $t_{44}^{0.05/50} = 3.2860719$ given significance level 0.05. We reject H_o whenever any of the Jacknife residual exceeds 3.2860719.

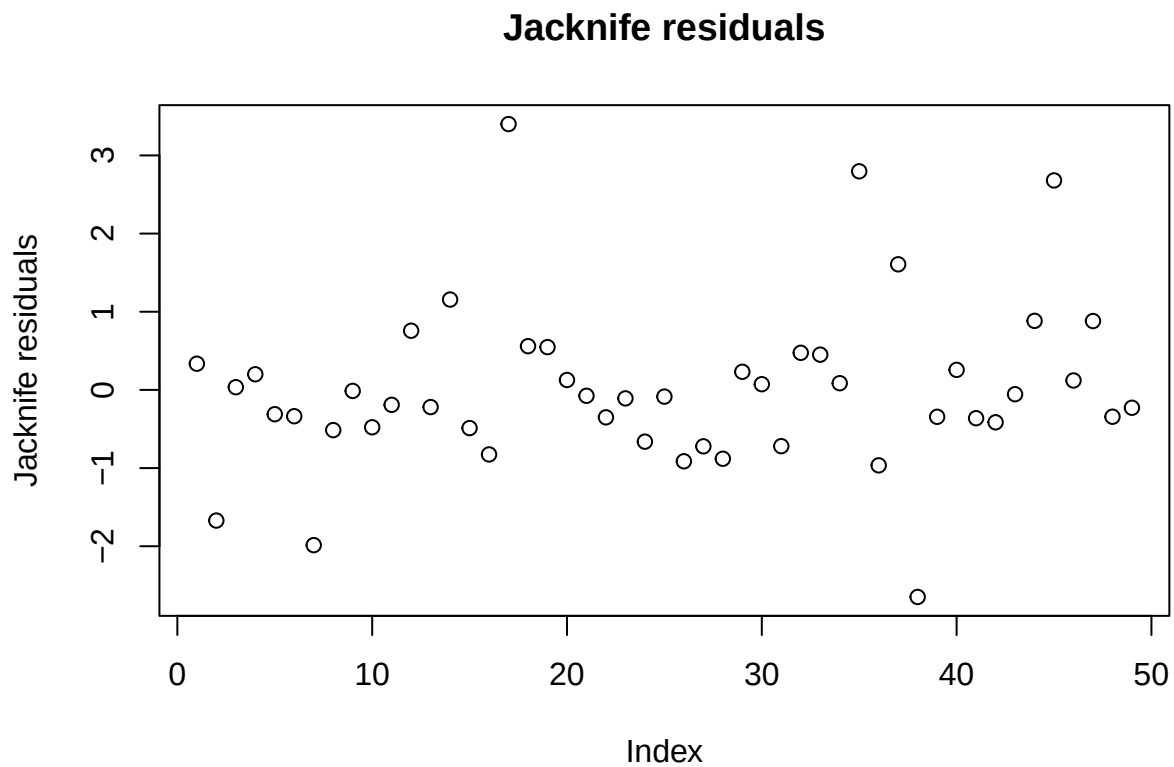
```r
jack[abs(jack) > critical]
```

```
##        9
## 5.255204
```

It turns out that the 9th observation whose Jacknife residual is 5.2552042 exceeds the critical value, and thus we conclude that there might be indeed at least one outlier in the data with respect to the working model. Moreover, the 9th observation is exactly the one. Therefore, we decide to remove the 9th observation and refit the model.

```
dat1 = dat1[-c(9),]
fit1 = lm(I(100*(Y84-Y83)/Y83)~., data = dat1)
```

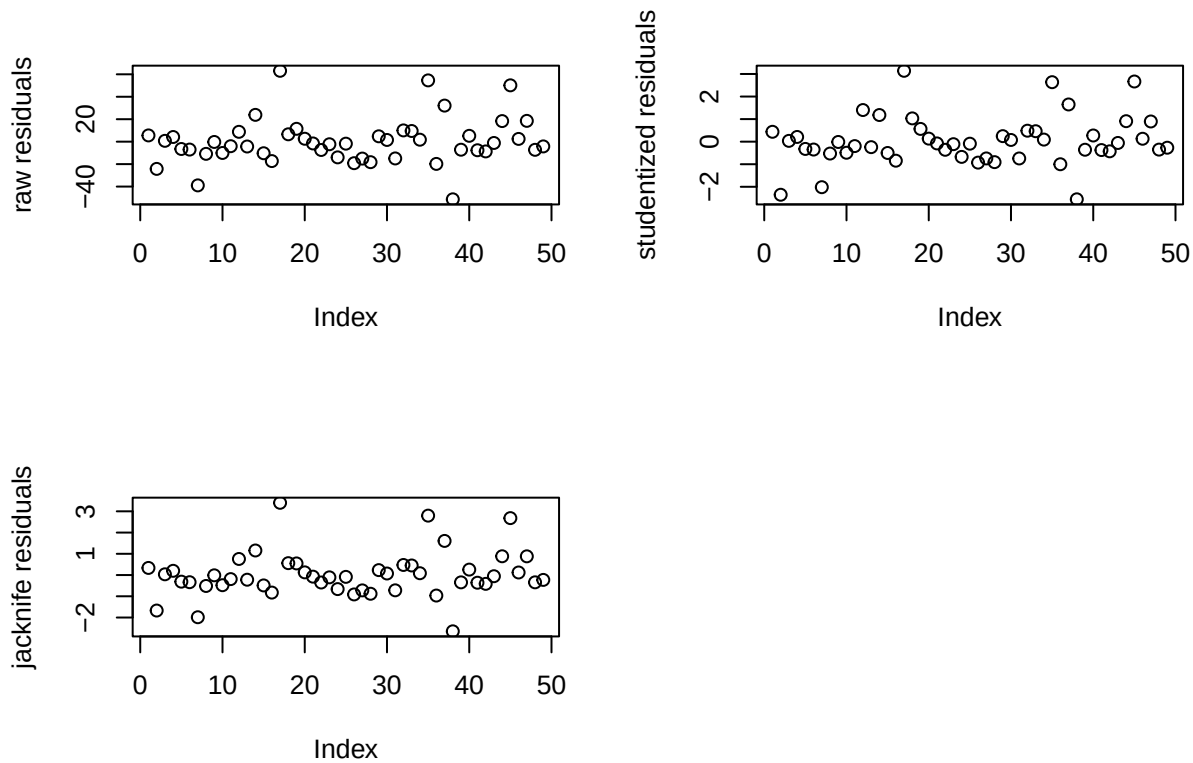
```
jack = rstudent(fit1)
plot(jack, ylab="Jacknife residuals", main="Jacknife residuals")
```



From the scatter plot above, we can see that there seems no sample points looking too strange any more.

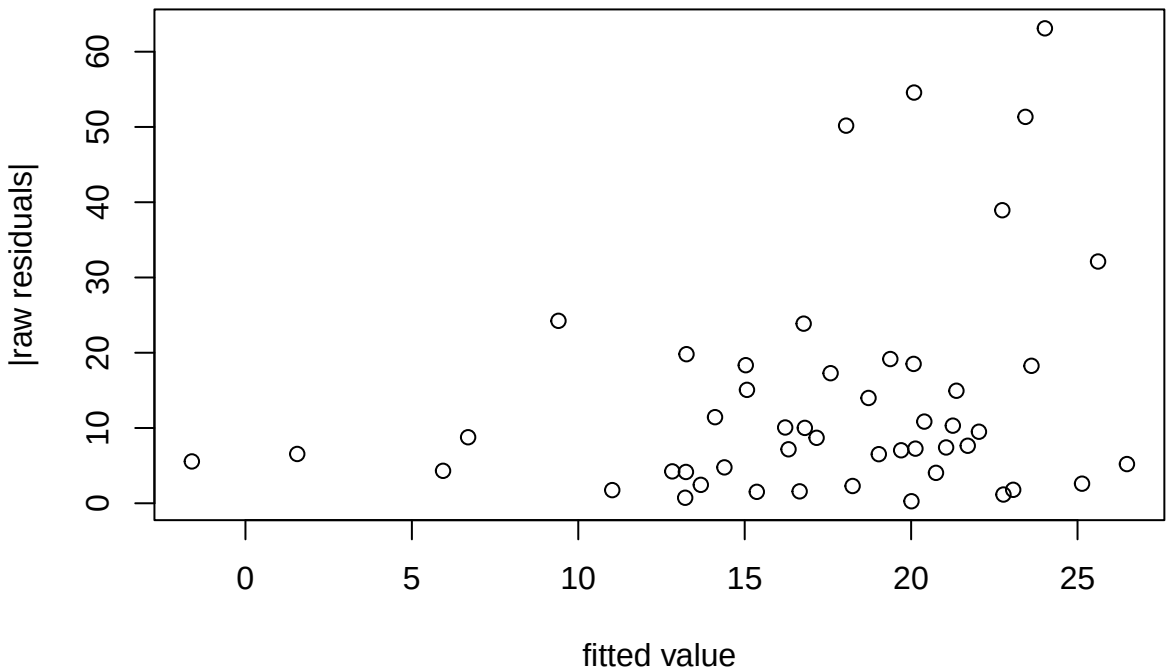
Examine nonconstant variance

```
lev = lm.influence(fit1)$hat
rstud = fit1$res/(summary(fit1)$sigma*(1-lev))
par(mfrow = c(2, 2))
plot(fit1$res,ylab="raw residuals")
plot(rstud,ylab="studentized residuals")
plot(jack,ylab="jacknife residuals")
```



The figure displayed above shows that the patterns of the three kinds of residual plots look similar. Therefore, We decide to use the raw residuals in the following analysis. To increase the resolution of detecting non-constant variance, the plot of absolute residual against fitted value is shown as below.

```
plot(abs(fit1$res)~fit1$fitted.values, ylab="|raw residuals|", xlab = "fitted value")
```



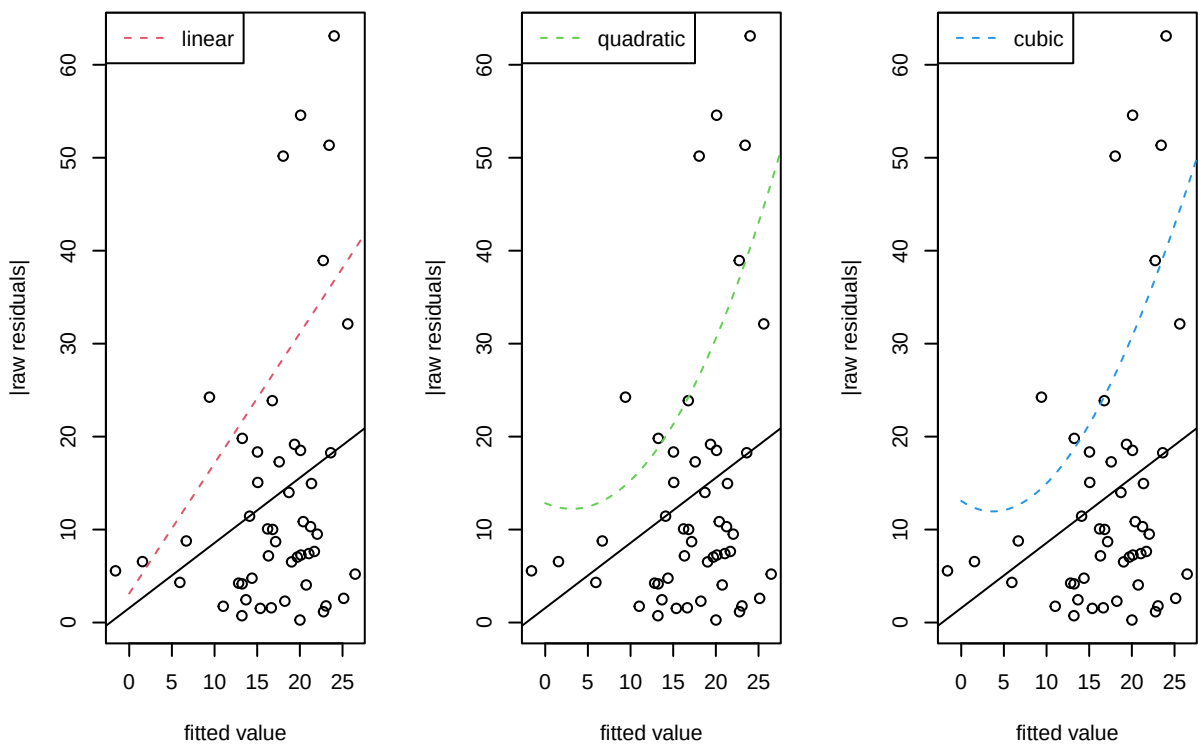
It seems that there might be problem of nonconstant variacne. We can roughly make a quick ckeck for it by regressing absolute residual on fitted value. The fitted result for the regression just described is shown as below, where we can see that the coefficient of fitted value is almost significant at 0.05 significance level. Such result suggests there might be problem of nonconstant variance.

```
summary(lm(abs(fit1$res)~fit1$fitted.values))
```

```
##
## Call:
## lm(formula = abs(fit1$res) ~ fit1$fitted.values)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.564  -8.695  -4.881   3.903  44.720
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.5597     6.5102   0.240   0.8117
## fit1$fitted.values   0.7004     0.3560   1.967   0.0551 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.48 on 47 degrees of freedom
## Multiple R-squared:  0.07608,    Adjusted R-squared:  0.05642
## F-statistic:  3.87 on 1 and 47 DF,  p-value: 0.05507
```

To investigate the relationship between the variance and mean, we try orthogonal polynomial regression of degree 1 to 3 with absolute residual as the response and fitted value as the predictor. Under such framework, the fitted absolute residual will be viewed as a surrogate for the standard deviation under different mean, which will be denoted by $\hat{\sigma}_{\text{fitted value}}$ in the following.

```
# plot(fit1$res~fit1$fitted.values, ylab="raw residuals", xlab = "fitted value")
par(mfrow = c(1, 3))
for(i in 1:3){
  plot(abs(fit1$res)~fit1$fitted.values, ylab="|raw residuals|", xlab = "fitted value")
  x = fit1$fitted.values
  fit = lm(abs(fit1$res)~poly(x, i))
  f = function(x){2*predict(fit, newdata = data.frame(x = x))}
  curve(f, from = 0, to = 30, col = i+1, lty = 2, add=TRUE)
  abline(lm(abs(fit1$res)~fit1$fitted.values))
  legend("topleft", c("linear", "quadratic", "cubic")[i], col = i+1, lty = 2)
}
```



After fitting those regression we just described, we plot the curves $2\hat{\sigma}_{\text{fitted value}}$ respectively on to the residual plots and see if most of the data points (should be approximately 95% of them) lie below the curves. Figure **??** shows the result and suggests that the standard deviation might be proportional to the quadratic mean (or cubic mean).

```
diff(range(100*(dat1$Y84-dat1$Y83)/dat1$Y83))
```

```
## [1] 115.0294
```

It can be shown that a suitable transformation under such situation is $g(y) = 1/y$. However, before taking this transformation, we should note that the 28th observation has response 0, which lets the transformation not applicable. To solve such problem, consider a constant shift 1. It is shown above that the range of the response is 115.0294. Therefore, shifting the response by 1 should be an acceptable measure since it would not make so much change.

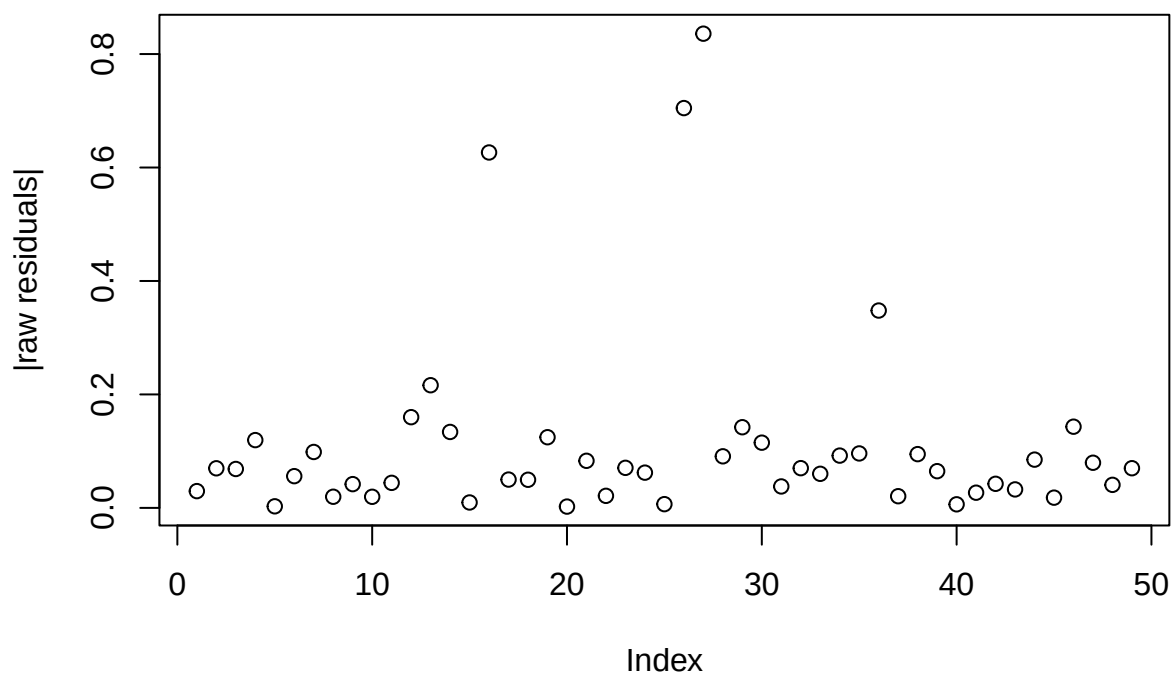
Transformation to improve matters

As discussed in the last section, we guess that transformation g will give constant variance That is, we consider an alternative model as follows.

Model: $g\left(\frac{100(Y84-Y83)}{Y83}+1\right) = \beta_0 + \beta_1 \text{SHARES} + \beta_2 \text{REV} + \beta_3 \text{INC} + \beta_4 \text{AGE} + \epsilon,\ \epsilon \sim N(0, \sigma^2 I)$

```
g = function(x){1/x}
fit2 = lm(I(g(100*(Y84-Y83)/Y83+1))~., data = dat1)
```

```
plot(abs(fit2$res), ylab="|raw residuals|")
```



From the residual plots displayed above, we see that some outliers seem to appear with respect to this new wording model. Hence, the same t-test which we have introduced in the previous section is conducted to detect those outliers.

```
jack = rstudent(fit2)
```

```
n = nrow(dat1); p = 5
critical = qt(.05/n, n-p-1, lower.tail = FALSE)
```

As what we have done before, we have in this time the critical value being $t_{43}^{0.05/49} = 3.2837536$ given significance level 0.05. We reject H_o whenever any of the Jacknife residual exceeds 3.2837536.

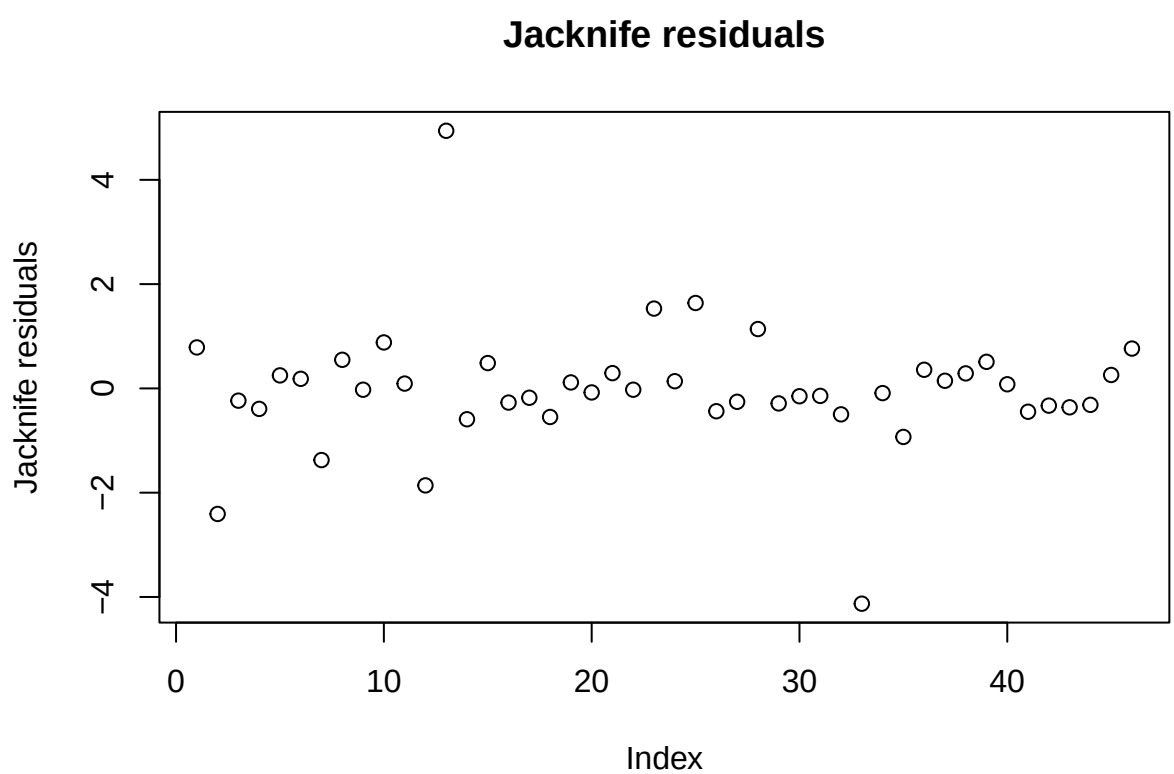
```
jack[abs(jack) > critical]
```

```
##       17       27       28
## 3.310385 3.835411 4.971492
```

It turns out that the 17th, 27th and 28th observation have their Jacknife residuals greater than the critical value, and thus we conclude that there might be indeed at least one outlier in the data with respect to the working model. Actually, the 17th, 27th and 28th observation are exactly the ones. Therefore, we decide to remove them from the data and refit the model.

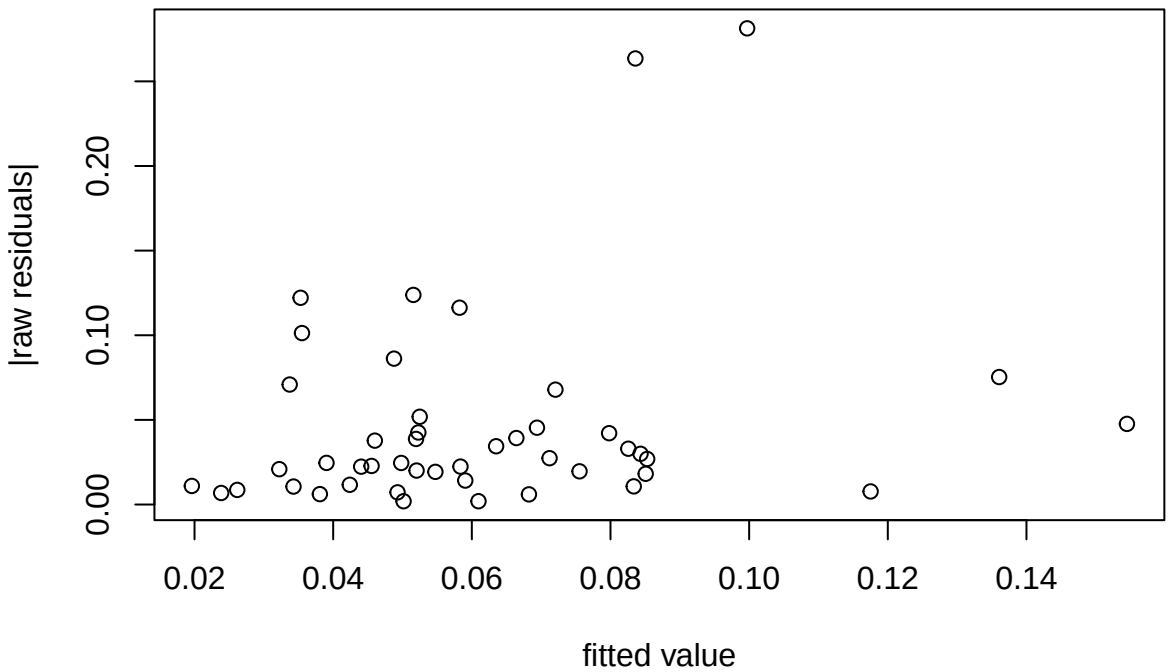
```
dat1 = dat1[-c(16, 26, 27),]
fit3 = lm(I(g(100*(Y84-Y83)/Y83+1))~., data = dat1)
```

```
jack = rstudent(fit3)
plot(jack, ylab="Jacknife residuals", main="Jacknife residuals")
```



From the scatter plot above, we can see that there seems no sample points looking too strange any more.

```
plot(abs(fit3$res)~fit3$fitted.values, ylab="|raw residuals|", xlab = "fitted value")
```



Now, to check whether the poblem of nonconstant variance has been fixed, we regress again the absolute residual on fitted value. The fitted summary for the regression just described is shown as below, where we can see that the coefficient of fitted value is quite apart from significant. Hence, we conclude that it becomes no strong evidence for problem of nonconstant variance. Such result is consistent with the residual plot displayed above, where the dispersion of absolute residual looks similar across the fitted values.

```
summary(lm(abs(fit3$res)~fit3$fitted.values))
```

```
##
## Call:
## lm(formula = abs(fit3$res) ~ fit3$fitted.values)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.064609 -0.028626 -0.016137 -0.001732  0.217271
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          0.01760    0.02086   0.844    0.403
## fit3$fitted.values  0.46574    0.31059   1.500    0.141
##
## Residual standard error: 0.05744 on 44 degrees of freedom
## Multiple R-squared:  0.04862,    Adjusted R-squared:  0.027
## F-statistic: 2.249 on 1 and 44 DF,  p-value: 0.1409
```

Summary for the final fit:

```r
summary(fit3)
```

```
##
## Call:
## lm(formula = I(g(100 * (Y84 - Y83)/Y83 + 1)) ~ ., data = dat1)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.263531 -0.026332 -0.003975  0.022682  0.281315
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.115e-01  1.050e-01  -1.061    0.295
## SHARES       6.487e-09  1.087e-08   0.597    0.554
## REV          1.426e-06  2.268e-06   0.629    0.533
## INC          1.773e-05  4.901e-05   0.362    0.719
## AGE          2.833e-03  1.841e-03   1.539    0.131
##
## Residual standard error: 0.0782 on 41 degrees of freedom
## Multiple R-squared:   0.12,  Adjusted R-squared:  0.03418
## F-statistic: 1.398 on 4 and 41 DF,  p-value: 0.2516
```

2 Q2

2.1 Fit a model with the octane rating as the response and the other four variables as predictors. Perform a full set of regression diagnostics. Present the important plots, comment on their meaning and, where appropriate, indicate what action should be taken. Since it is easy to produce a very large number of plots, you will need to be selective. Take care to present examples of each major category of diagnostic. It is acceptable to just report the outcome of some plots without actually displaying them, especially if they do not show anything interesting.

Model: $rating = \beta_0 + \beta_1\, A1 + \beta_2\, A2 + \beta_3\, A3 + \beta_4\, A4 + \epsilon,\ \epsilon \sim N(0, \sigma^2 I)$

```r
dat2 = read.table("octane.txt")
fit3 = lm(rating~., data = dat2)
summary(fit3)
```

```
##
## Call:
## lm(formula = rating ~ ., data = dat2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00612 -0.28588 -0.04679  0.32159  0.98069
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 95.853150   1.224877  78.255  < 2e-16 ***
```

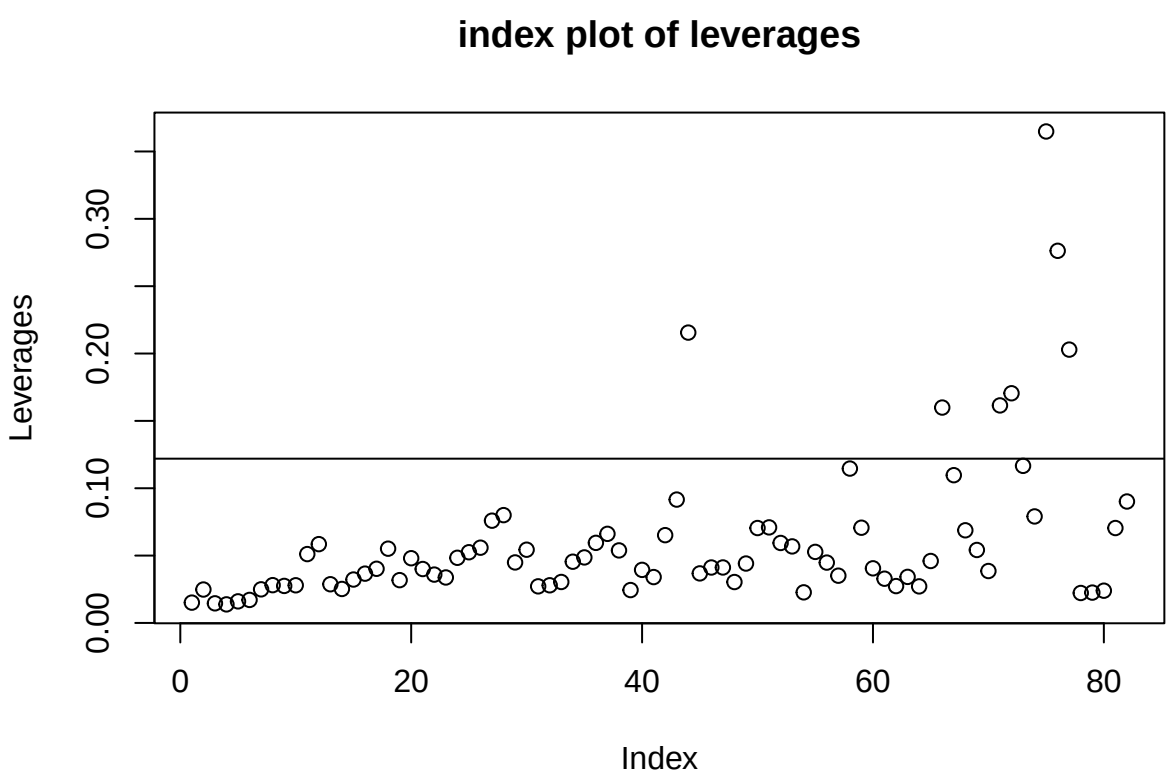
```
## A1          -0.092821   0.005235 -17.729  < 2e-16 ***
## A2          -0.126798   0.032157  -3.943 0.000176 ***
## A3          -0.025381   0.013971  -1.817 0.073160 .
## A4           1.967603   0.324573   6.062 4.65e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4415 on 77 degrees of freedom
## Multiple R-squared:  0.9056, Adjusted R-squared:  0.9007
## F-statistic: 184.7 on 4 and 77 DF,  p-value: < 2.2e-16
```

```r
n = nrow(dat2)
p = 5
```

The sample size n = 82, and the number of parameters p = 5.

Diagnostic for leverage point

```r
lev = lm.influence(fit3)$hat
plot(lev, ylab = "Leverages", main = "index plot of leverages")
abline(h = 2*p/n)
```



The scatter plot above shows the leverage of each observation. A horizontal line at 2p/n, which is decided by the rule of thumb, is marked on the plot to help identify those with too large leverage.

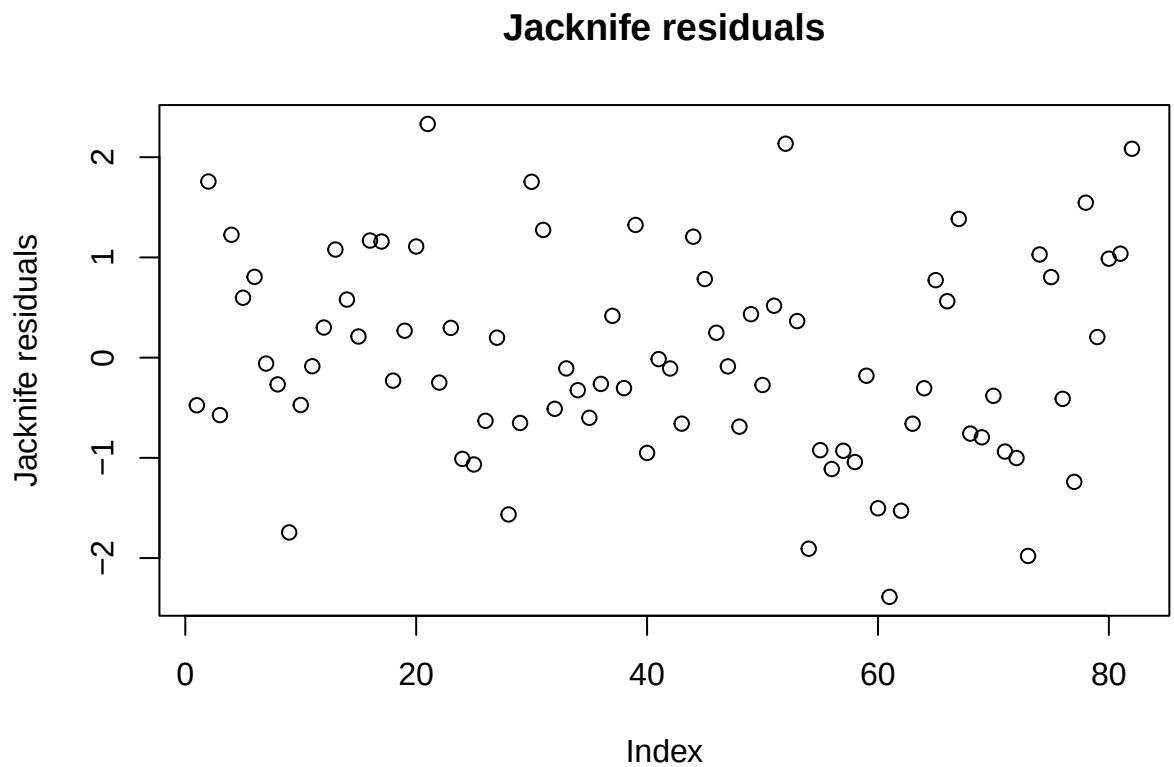
```r
lev[lev > 2*p/n]
```

```
##        44        66        71        72        75        76        77
## 0.2155490 0.1599043 0.1614875 0.1705040 0.3648473 0.2762653 0.2029067
```

It is found that the 44th, 75th, 76th and 77th observation seem to have too large leverage, and we thus identify them as the leverage points.

Diagnostic for outlier

```r
jack = rstudent(fit3)
plot(jack, ylab="Jacknife residuals", main="Jacknife residuals")
```



From the scatter plot above, it seems that none of the observations look like outlier. However, it is still better to go through some numerical approach for more objective checking.

It can be shown that Jacknife residuals follow t distribution with degree of freedom $n-p-1$ under null if the model is correct and $\epsilon \sim N(0, \sigma^2 I)$. Hence hypothesis testing described as follows can be conducted to examine if their is any outlier present in this data.

$$\begin{cases} H_0 : \text{No any outlier.} \\ H_1 : \text{There is at least one outlier.} \end{cases}$$

```
critical = qt(.05/n, n-p-1, lower.tail = FALSE)
```

Note that this proposed test is involved with multiple testing. Bonferroni's correction is one of the measures to deal with such problem. Since the sample size n = 82 and the number of parameters p = 5, we have the critical value for this test being $t_{76}^{0.05/82} = 3.3605022$ given significance level 0.05. We reject H_o whenever any of the Jacknife residual exceeds 3.3605022.

```
jack[abs(jack) > critical]
```

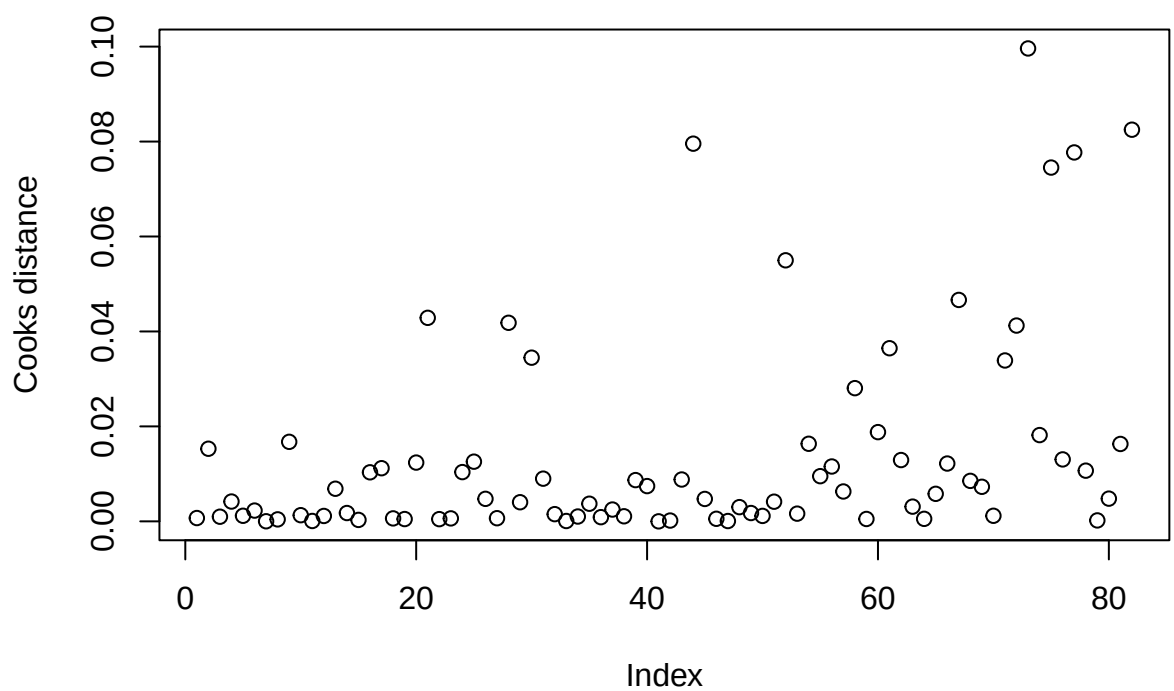
```
## named numeric(0)
```

It turns out that none of the residuals exceed the critical value, and we thus conclude that we do not have strong evidence to say there is any outlier present in the data. This result is consistent with our conjecture when looking at the residual plot.

Diagnostic for influential point

Cook's distance is used to examine whether there is any influential point.

```
cook = cooks.distance(fit3)
plot(cook, ylab = "Cooks distance")
```



The plot displayed above shows that some of the observations might have too large Cook's distance. Such observations seem to be influential point.

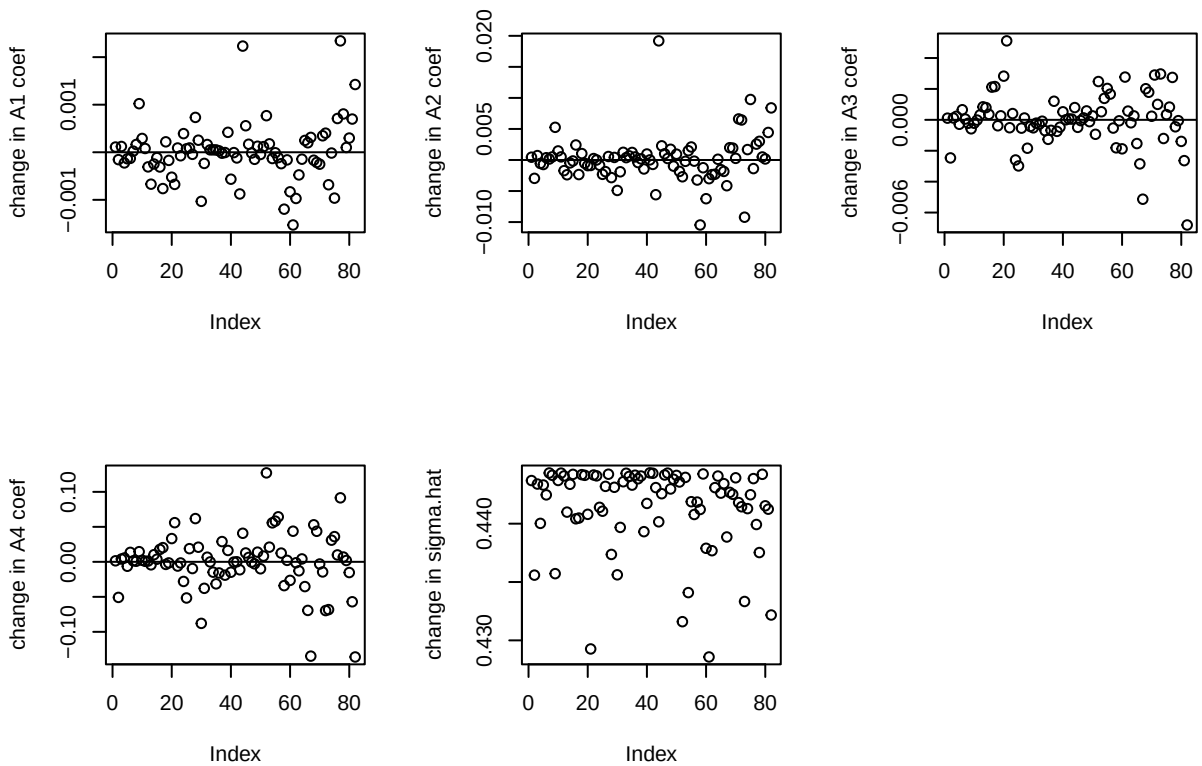
```r
names(sort(cook, decreasing = TRUE)[1:3])
```

```
## [1] "73" "82" "44"
```

It is discovered that the 13rd, 2nd and 14th observation have the most influence on the fitting result. We decide to look more deeply to see if there is still any other observation which would have strong influence on the estimation for some specific coefficients.

```r
fit3inf = lm.influence(fit3)
```

```r
inf.ls = list()
par(mfrow = c(2, 3))
for(i in 1:4){
  plot(fit3inf$coef[,i+1], ylab = paste("change in A", i, " coef", sep = "")); abline(h = 0)
  inf.ls[[i]] = names(sort(fit3inf$coef[,i+1], decreasing = TRUE)[1:3])
}
plot(fit3inf$sigma, ylab = "change in sigma.hat"); abline(h = 0)
inf.ls[[5]] = names(sort(fit3inf$sigma, decreasing = TRUE)[1:3])
```



The figure above shows how each coefficient and $\hat{\sigma}$ would be changed when fitting model without using each of the observation. The larger the change, the higher the influence an observation has on the corresponding parameter estimate. We see that there indeed seems to be some influential points for each parameter estimate. Table 1 summarizes which observation might have influence on each parameter estimate, where "*" denotes that the corresponding observation is among the first three most influential points for the corresponding parameter estimate.

Table 1: Summary for the influential points

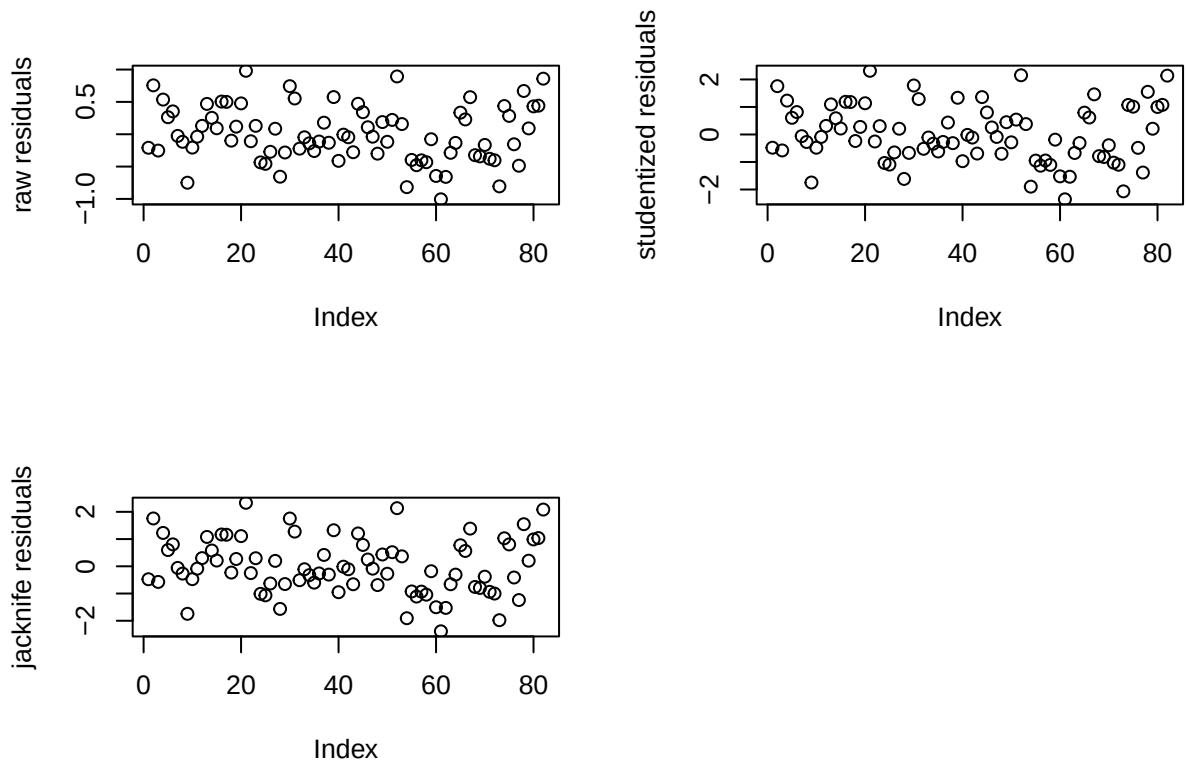
	77	44	82	75	21	73	71	52	56	41	7	11
A1	*	*	*									
A2		*	*	*								
A3					*	*	*					
A4	*							*	*			
$\hat{\sigma}$										*	*	*

```
summ = matrix("", ncol = length(unique(unlist(inf.ls))), nrow = 5)
colnames(summ) = unique(unlist(inf.ls))
rownames(summ) = c("A1", "A2", "A3", "A4", "$\\hat{\\sigma}$")
for(i in 1:5){
  fill = rep("", 12)
  fill[unique(unlist(inf.ls)) %in% inf.ls[[i]]] = "*"
  summ[i,] = fill
}
kable(summ, caption = "Summary for the influential points",
      escape = FALSE, booktabs = TRUE)
```

Though there seems to be some sample points having "relatively" large influence, we choose not to remove any of them since the magnitude of their influence does not seems to be really big when comparing to the coefficient estimates obtained with the full data.

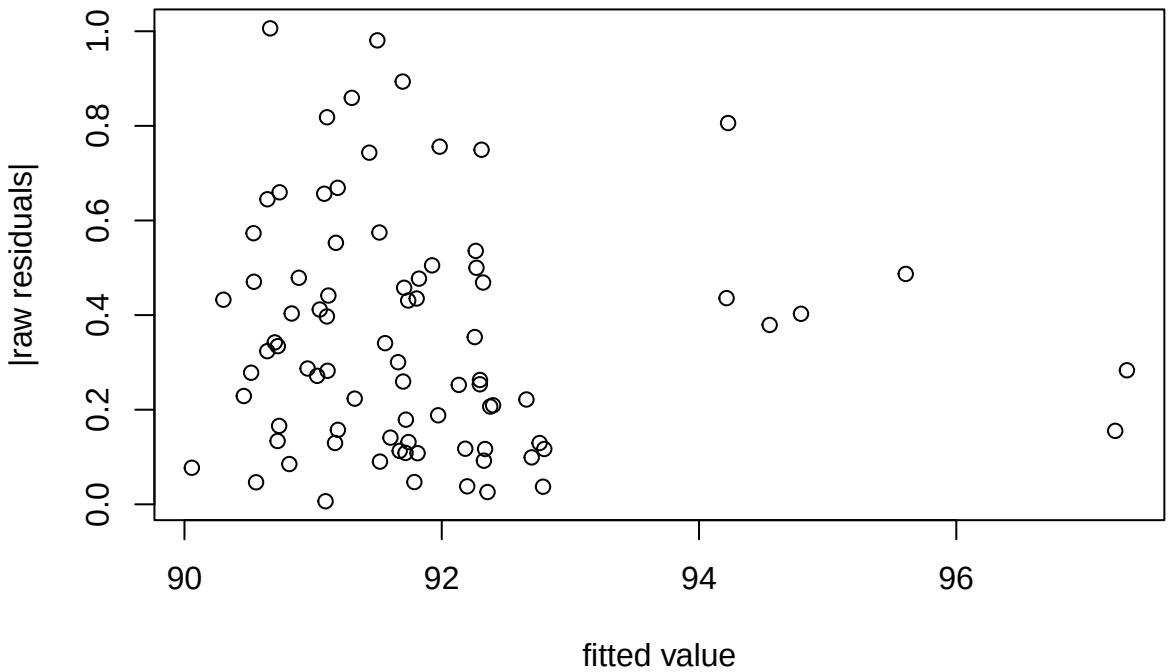
Diagnostic for nonconstant variance

```
lev = lm.influence(fit3)$hat
rstud = fit3$res/(summary(fit3)$sigma*(1-lev))
par(mfrow = c(2, 2))
plot(fit3$res,ylab="raw residuals")
plot(rstud,ylab="studentized residuals")
plot(jack,ylab="jacknife residuals")
```



The figure displayed above shows that the patterns of the three kinds of residual plots look similar. Therefore, We decide to use the raw residuals in the following analysis. To increase the resolution of detecting non-constant variance, the plot of absolute residual against fitted value is shown as below.

```
plot(abs(fit3$res)~fit3$fitted.values, ylab="|raw residuals|", xlab = "fitted value")
```



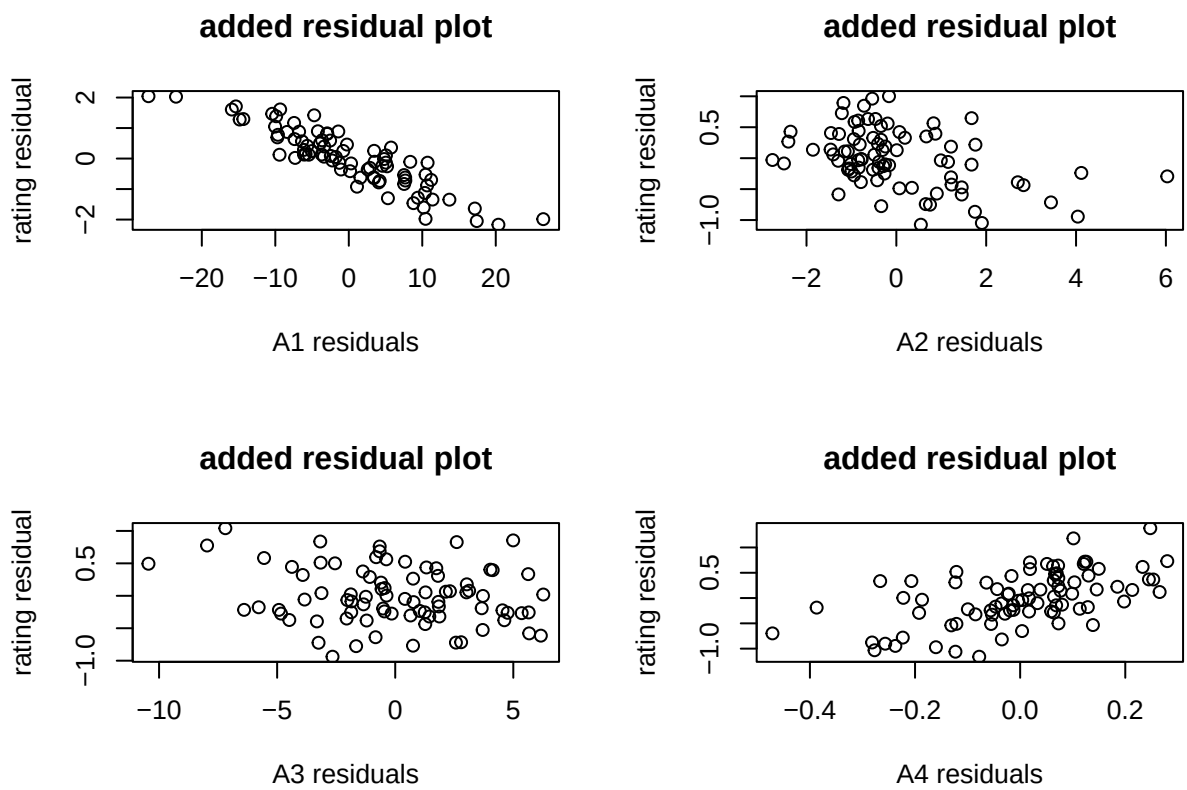
It seems that there might not be problem of nonconstant variacne. We can roughly make a quick ckeck for it by regressing absolute residual on fitted value. The fitted result for the regression just described is shown as below, where we can see that the coefficient of fitted value is far apart from significant at 0.05 significance level. Hence, we conclude that the assumption for constant variance might be acceptable.

```
summary(lm(abs(fit3$res)~fit3$fitted.values))
```

```
##
## Call:
## lm(formula = abs(fit3$res) ~ fit3$fitted.values)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.35569 -0.21312 -0.04933  0.12456  0.63802
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.61887    1.88125   0.861    0.392
## fit3$fitted.values -0.01380    0.02048  -0.674    0.502
##
## Residual standard error: 0.2458 on 80 degrees of freedom
## Multiple R-squared:  0.00564,    Adjusted R-squared:  -0.00679
## F-statistic: 0.4538 on 1 and 80 DF,  p-value: 0.5025
```

Diagnostic for curvature in the mean of residuals

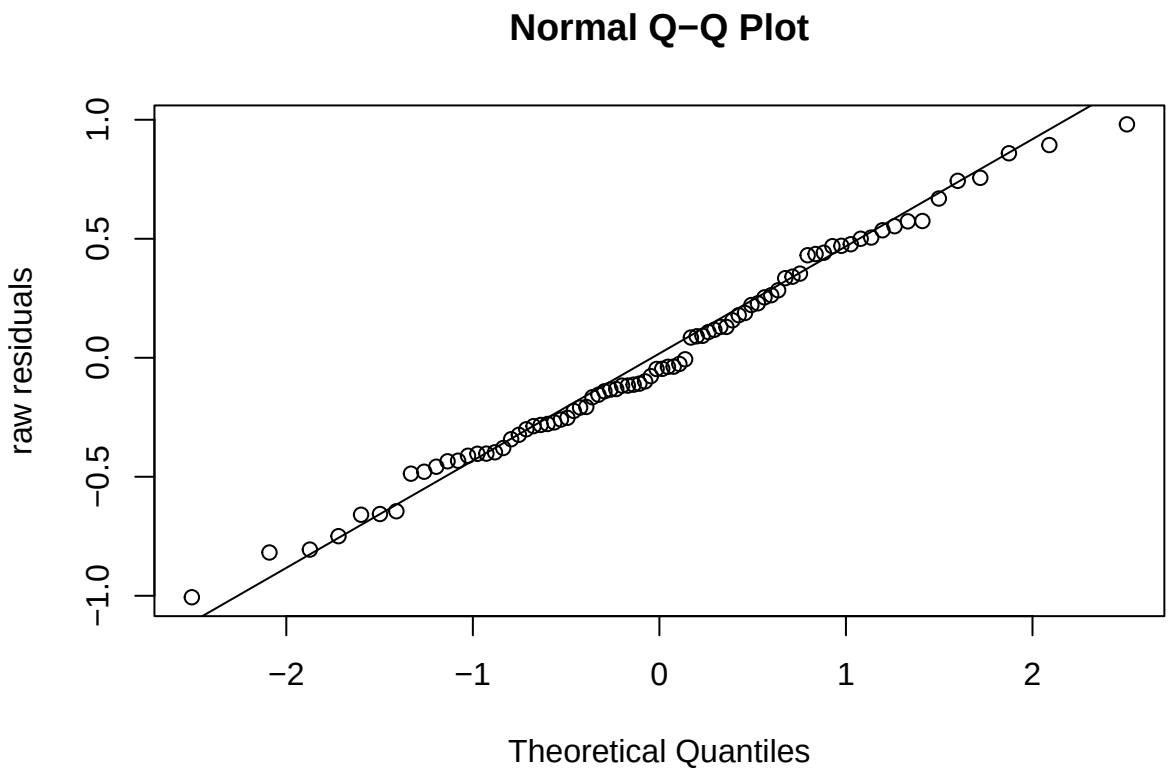
```r
par(mfrow = c(2, 2))

d = lm(rating~A2+A3+A4, data = dat2)$res
m = lm(A1~A2+A3+A4, data = dat2)$res
plot(m, d, xlab = "A1 residuals", ylab = "rating residual", main = "added residual plot")

d = lm(rating~A1+A3+A4, data = dat2)$res
m = lm(A2~A1+A3+A4, data = dat2)$res
plot(m, d, xlab = "A2 residuals", ylab = "rating residual", main = "added residual plot")

d = lm(rating~A1+A2+A4, data = dat2)$res
m = lm(A3~A1+A2+A4, data = dat2)$res
plot(m, d, xlab = "A3 residuals", ylab = "rating residual", main = "added residual plot")

d = lm(rating~A1+A2+A3, data = dat2)$res
m = lm(A4~A1+A2+A3, data = dat2)$res
plot(m, d, xlab = "A4 residuals", ylab = "rating residual", main = "added residual plot")
```



Added residual plots are displayed above for each predictor. Since none of the plots show relationship between the two kinds of residuals other than linear relation, we conclude that there might not be curvature in the mean of residuals.

Diagnostic for Normality

```r
qqnorm(fit3$res, ylab = "raw residuals")
qqline(fit3$res)
```



Q-Q plot is used to check for normality. Since all the points lie quite close to the line joining the first and third quartiles, we conclude that there is no strong evidence to say that the normality assumption does not hold. In other words, normality might be an acceptable assumption for this data.

Diagnostic for correlated errors

Since there is no information about time index included in the model, we can not check for correlated errors.

3 Q3

3.1 The data for this question gives observations on the acceleration (ACC) of different vehicles along with their weight-to-horsepower ration (WHP), the speed at which they were traveling (SP), and the grade (G; G=0 implies the road was horizontal).

a. Run a regression using ACC as your dependent variable without making any transformations, and obtain the partial residual plots

Read data:

```
dat3 = read.table("vehicle.txt", header = TRUE)
```

Model: $ACC = \beta_0 + \beta_1 WHP + \beta_2 SP + \beta_3 G + \epsilon$, $\epsilon \sim N(0, \sigma^2 I)$

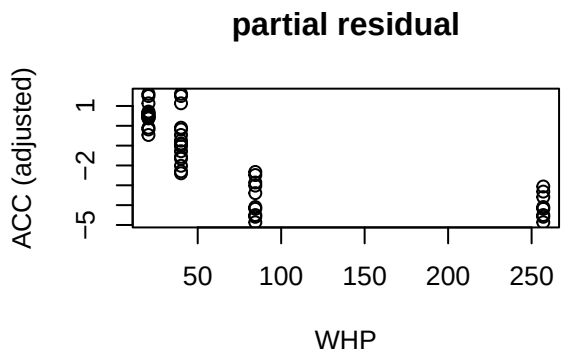
Summary for the fit:

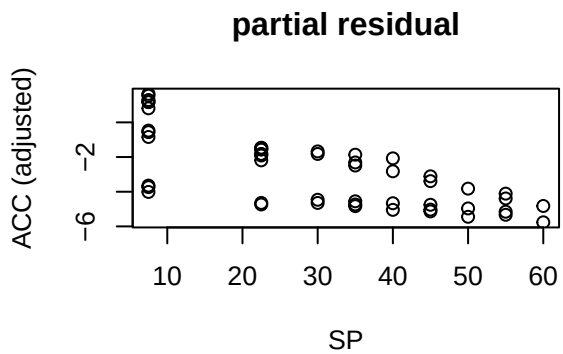
```r
fit4 = lm(ACC~., data = dat3)
summary(fit4)
```

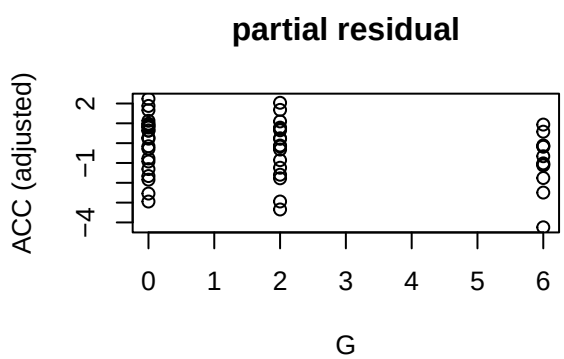
```
##
## Call:
## lm(formula = ACC ~ ., data = dat3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3124 -0.9003  0.2486  0.9489  2.3477
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.19949    0.60087  11.982 9.57e-16 ***
## WHP         -0.01838    0.00269  -6.833 1.62e-08 ***
## SP          -0.09347    0.01307  -7.149 5.45e-09 ***
## G           -0.15548    0.09040  -1.720   0.0922 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.47 on 46 degrees of freedom
## Multiple R-squared:  0.624,  Adjusted R-squared:  0.5995
## F-statistic: 25.45 on 3 and 46 DF,  p-value: 7.451e-10
```

Partial residual plots:

```r
par(mfrow = c(2, 2))
plot(dat3$WHP, fit4$res+fit4$coef["WHP"]*dat3$WHP, xlab = "WHP",
     ylab = "ACC (adjusted)", main = "partial residual")
plot(dat3$SP, fit4$res+fit4$coef["SP"]*dat3$SP, xlab = "SP",
     ylab = "ACC (adjusted)", main = "partial residual")
plot(dat3$G, fit4$res+fit4$coef["G"]*dat3$G, xlab = "G",
     ylab = "ACC (adjusted)", main = "partial residual")
```







According to the partial residual plots displayed above, It is found that there might be curvature relation between WHP and ACC and SP and ACC. Furthermore, one should note that the partial residual plot for SP also suggests problem of heteroscedasticity.

Obtain a good fitting model by making whatever changes you think are necessary. Obtain appropriate plots to verify that you have succeeded.

Since it is suspected that there might be curvature relation between WHP and ACC and SP and ACC, we rewrite the model as follows.

Model: $ACC = \beta_0 + \beta_1 WHP + \beta_{11} WHP^2 + \beta_2 SP + \beta_{22} SP^2 + \beta_3 G + \epsilon$

Summary for the fit:

```
fit4 = lm(ACC~poly(WHP, 2)+poly(SP, 2)+G, data = dat3)
summary(fit4)
```

```
##
## Call:
## lm(formula = ACC ~ poly(WHP, 2) + poly(SP, 2) + G, data = dat3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.53246 -0.74447 -0.06437  0.58039  2.13169
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     3.09018    0.17612  17.546  < 2e-16 ***
```

```
## poly(WHP, 2)1 -10.88890    0.96605 -11.272 1.47e-14 ***
## poly(WHP, 2)2   7.90305    0.92353   8.557 6.48e-11 ***
## poly(SP, 2)1  -11.76665    0.95635 -12.304 7.71e-16 ***
## poly(SP, 2)2    1.14711    0.91894   1.248 0.218523
## G              -0.22839    0.05668  -4.030 0.000218 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9105 on 44 degrees of freedom
## Multiple R-squared:  0.862,  Adjusted R-squared:  0.8464
## F-statistic: 54.99 on 5 and 44 DF,  p-value: < 2.2e-16
```

The summary above shows that all the predictors are significant at 0.05 significant level except for the quadratic term of SP. Hence, we decide to remove SP^2 and refit the model as follows.

Model: $ACC = \beta_0 + \beta_1 WHP + \beta_{11} WHP^2 + \beta_2 SP + \beta_3 G + \epsilon$

Summary for the fit:

```r
fit4 = lm(ACC~poly(WHP, 2)+SP+G, data = dat3)
summary(fit4)
```

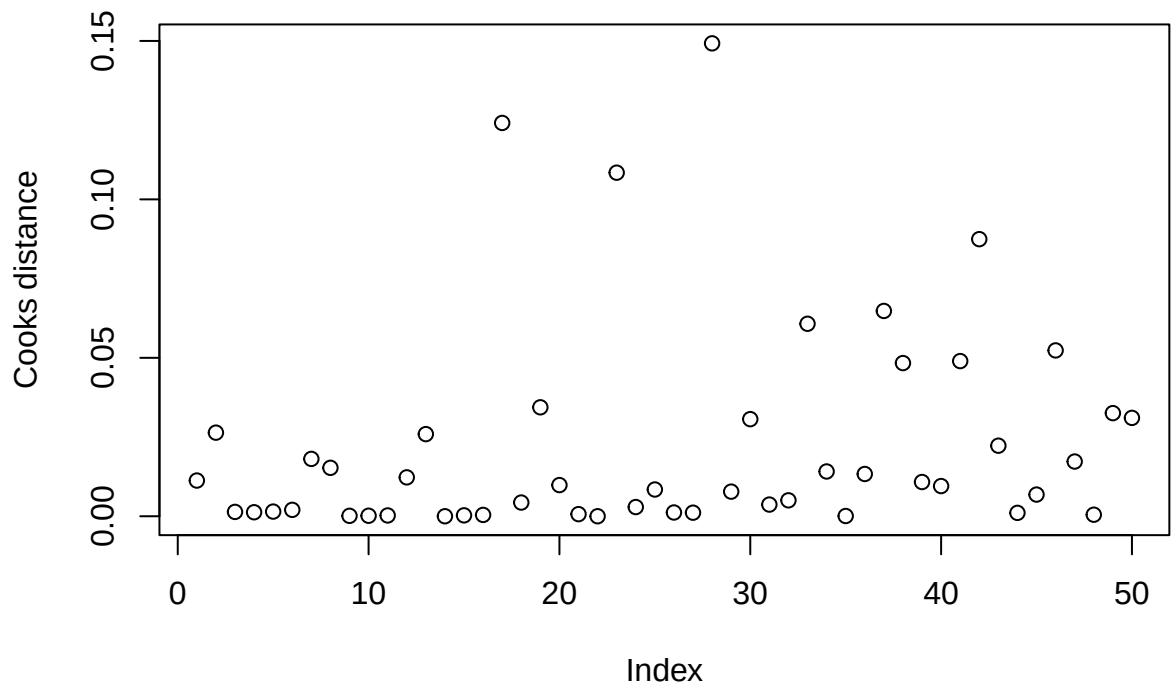
```
##
## Call:
## lm(formula = ACC ~ poly(WHP, 2) + SP + G, data = dat3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.70446 -0.64576 -0.05457  0.54006  2.28414
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     6.131465   0.318232  19.267  < 2e-16 ***
## poly(WHP, 2)1 -10.997445   0.968074 -11.360 8.25e-15 ***
## poly(WHP, 2)2   7.955503   0.928283   8.570 5.18e-11 ***
## SP             -0.100379   0.008188 -12.259 6.08e-16 ***
## G              -0.223561   0.056893  -3.929  0.00029 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9161 on 45 degrees of freedom
## Multiple R-squared:  0.8572, Adjusted R-squared:  0.8445
## F-statistic: 67.51 on 4 and 45 DF,  p-value: < 2.2e-16
```

To be more confident about our fitting result, it is better to conduct some model diagnostic. At this stage, we only check for the influential points.

Diagnostic for influential points

Cook's distance is used to examine whether there is any influential point.

```r
cook = cooks.distance(fit4)
plot(cook, ylab = "Cooks distance")
```



The plot displayed above shows that some of the observations might have too large Cook's distance. Such observations seem to be influential points.

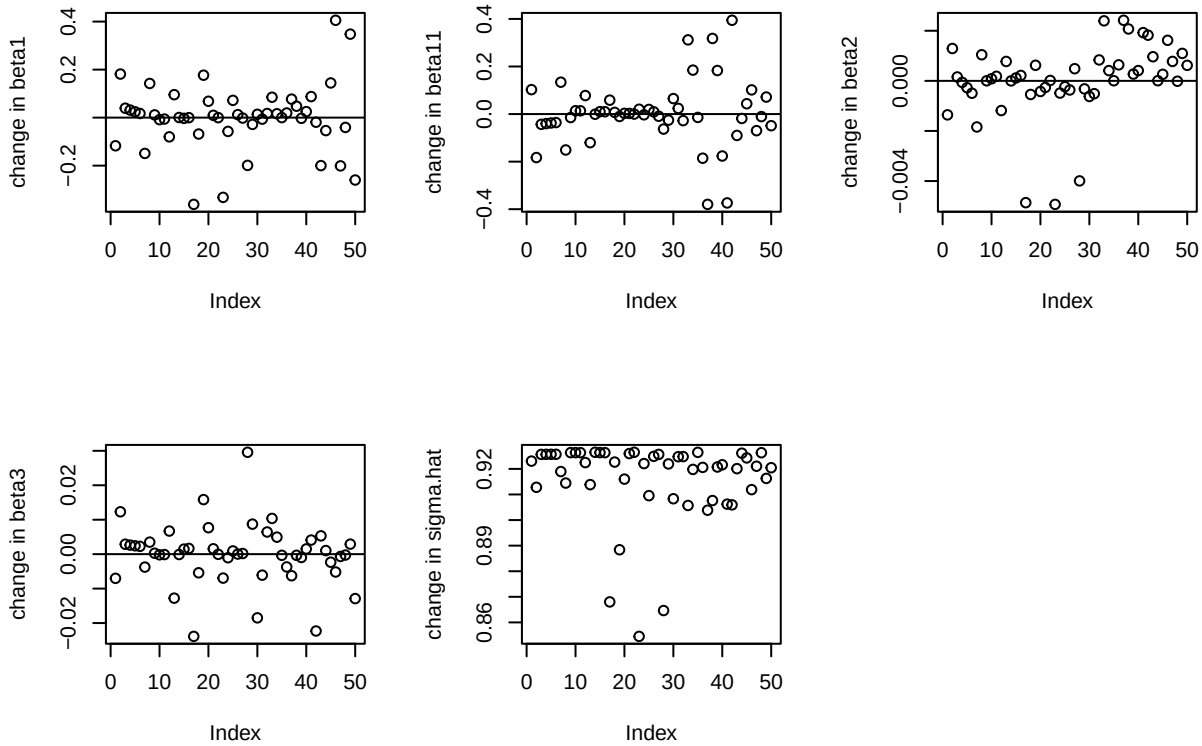
```
names(sort(cook, decreasing = TRUE)[1:3])
```

```
## [1] "28" "17" "23"
```

It is discovered that the 28th, 17th and 23rd observation have the most influence on the fitting result. We decide to look more deeply to see if there is still any other observation which would have strong influence on the estimation for some specific coefficients.

```
fit4inf = lm.influence(fit4)
```

```
inf.ls = list()
par(mfrow = c(2, 3))
cols = c("beta1", "beta11", "beta2", "beta3")
for(i in 1:4){
  plot(fit4inf$coef[,i+1], ylab = paste("change in",cols[i])); abline(h = 0)
  inf.ls[[i]] = names(sort(fit4inf$coef[,i+1], decreasing = TRUE)[1:3])
}
plot(fit4inf$sigma, ylab = "change in sigma.hat"); abline(h = 0)
inf.ls[[5]] = names(sort(fit4inf$sigma, decreasing = TRUE)[1:3])
```



The figure above shows how each coefficient and $\hat{\sigma}$ would be changed when fitting model without using each of the observation. The larger the change, the higher the influence an observation has on the corresponding parameter estimate. We see that there indeed seems to be some influential points for each parameter estimate. Table 2 summarizes which observation might have influence on each parameter estimate, where "*" denotes that the corresponding observation is among the first three most influential points for the corresponding parameter estimate.

```
summ = matrix("", ncol = length(unique(unlist(inf.ls))), nrow = 5)
colnames(summ) = unique(unlist(inf.ls))
rownames(summ) = c(cols, "$\\hat{\\sigma}$")
for(i in 1:5){
  fill = rep("", length(unique(unlist(inf.ls))))
  fill[unique(unlist(inf.ls)) %in% inf.ls[[i]]] = "*"
  summ[i,] = fill
}
kable(summ, caption = "Summary for the influential points",
      escape = FALSE, booktabs = TRUE)
```

Though there seems to be some sample points having "relatively" large influence, we choose not to remove any of them since the magnitude of their influence does not seems to be really big when comparing to the coefficient estimates obtained with the full data.

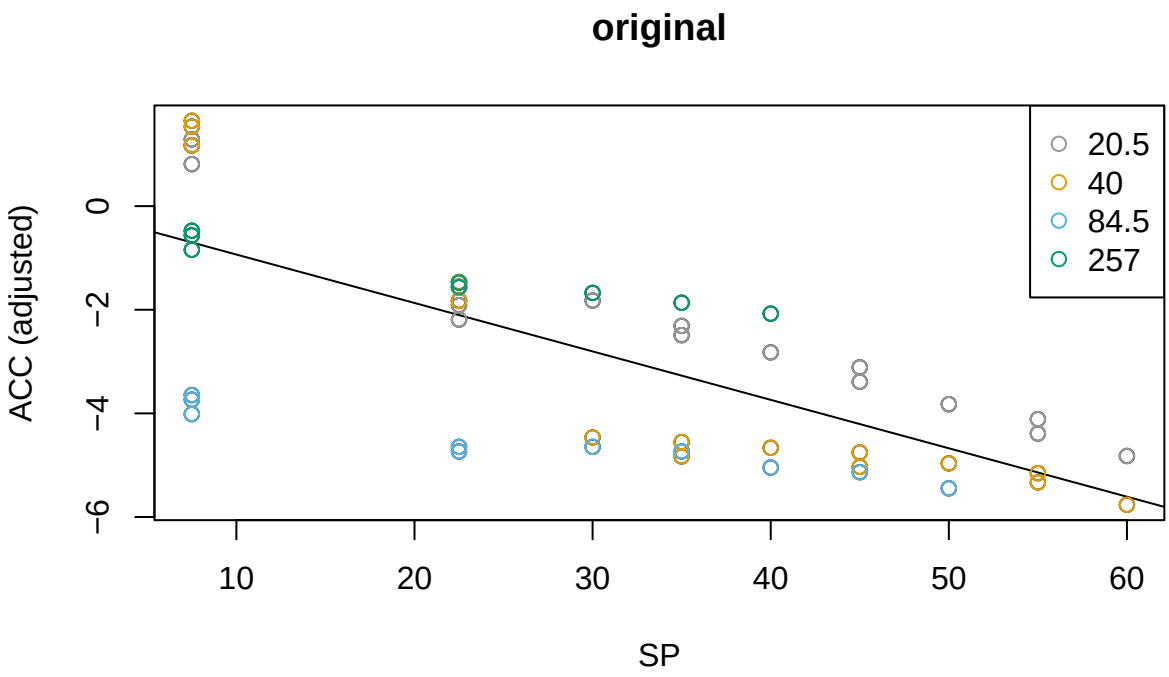
c. At least one of the partial residual plots in part a appears to show heteroscedascity (i.e., violation of var( )= 2I).. If you have been successful in part b, the appearance of any serious heteroscedascity should vanish without your having to weight or transform the dependent variable. Explain why you think this happens.

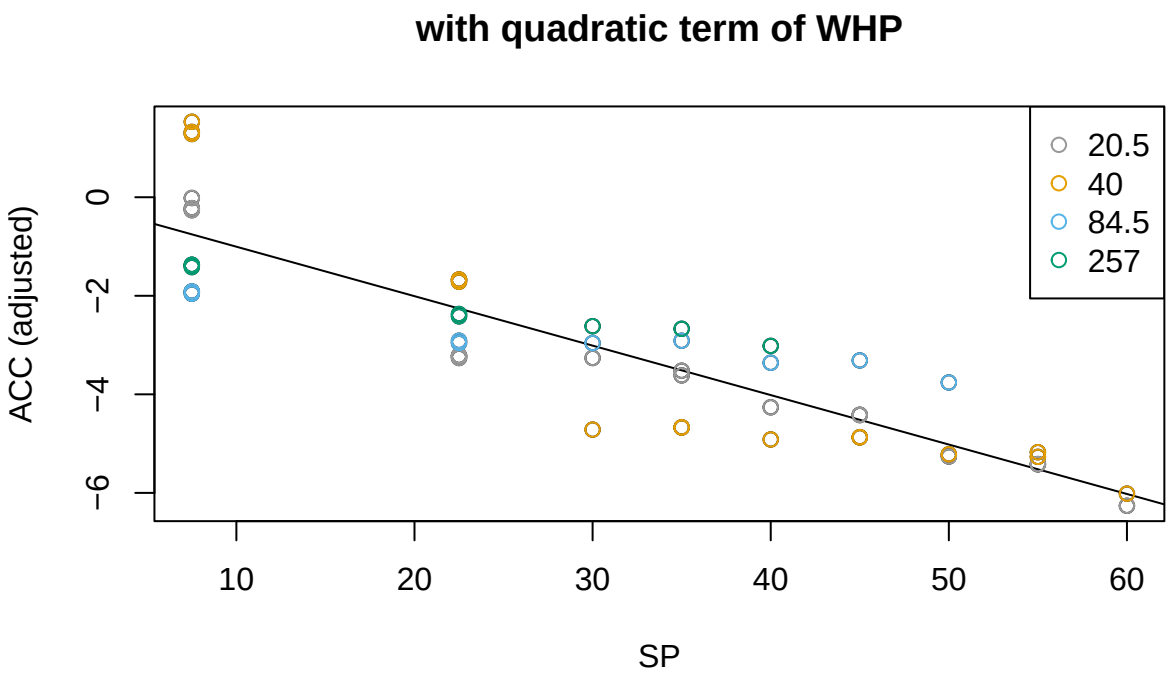
Table 2: Summary for the influential points

	46	49	2	42	38	33	37	28	19	22	14	35
beta1	*	*	*									
beta11				*	*	*						
beta2					*	*	*					
beta3			*					*	*			
$\hat{\sigma}$										*	*	*

We have mentioned the problem of heteroscedasticity with respect to SP in part a. Partial residual plots are plotted again for checking whether the heteroscedasticity has vanished after adding the quadratic term of WHP.

```
par(mfrow = c(2, 1))
cbp = c("#999999", "#E69F00", "#56B4E9", "#009E73",
        "#F0E442", "#0072B2", "#D55E00", "#CC79A7",
        "#00AFBB", "#E7B800", "#FC4E07","black")

fit4 = lm(ACC~., data = dat3)
plot(dat3$SP, fit4$res+fit4$coef["SP"]*dat3$SP, xlab = "SP",
     ylab = "ACC (adjusted)", main = "original")
abline(lm(fit4$res+fit4$coef["SP"]*dat3$SP~dat3$SP))
dat3$WHP = factor(dat3$WHP)
lv = levels(dat3$WHP)
for(i in 1:4){
  points(dat3$SP[dat3$WHP == lv[i]],
         (fit4$res+fit4$coef["SP"]*dat3$SP)[dat3$WHP == lv[i]], col = cbp[i])
}
legend("topright", lv, col = cbp[1:4], pch = 1)
dat3 = read.table("vehicle.txt", header = TRUE)


fit4 = lm(ACC~poly(WHP, 2)+SP+G, data = dat3)
plot(dat3$SP, fit4$res+fit4$coef["SP"]*dat3$SP, xlab = "SP",
     ylab = "ACC (adjusted)", main = "with quadratic term of WHP")
abline(lm(fit4$res+fit4$coef["SP"]*dat3$SP~dat3$SP))
dat3$WHP = factor(dat3$WHP)
lv = levels(dat3$WHP)
for(i in 1:4){
  points(dat3$SP[dat3$WHP == lv[i]],
         (fit4$res+fit4$coef["SP"]*dat3$SP)[dat3$WHP == lv[i]], col = cbp[i])
}
legend("topright", lv, col = cbp[1:4], pch = 1)
```





The two partial residual plots above show that the heteroscedasticity indeed vanished. In the figure above, we also plot the points by different color in correspondence with the WHP associated on each point. In the original partial residual plot, we can see that the most points lying below the black line are those associating with WHP value 40 and 84.5 (values closed to the mean of WHP), while the most points lying along the black line are those associating with WHP value 20.5 and 257 (values far apart from the mean of WHP). Since

we have found the quadratic effect of WHP (i.e., curvature pattern in the partial residual plot for WHP) according to the analysis just now, we can thereby explain why points with WHP near the mean point have relatively low residuals, while those with WHP far apart from the mean point have relatively high residuals. It is because such pattern is actually of curved nature! In other words, the reason why the heteroscedasticity vanished should be because that the quadratic effect of WHP reflect on the partial residual plot for SP, and hence such pattern would get vanished once we add the quadratic term of WHP.